

Parasites

Surprisingly, the world's biggest flower is produced by a plant with no leaves. Rafflesia's massive bloom stinks of rotting meat to attract pollinating blowflies, but the rest of the plant grows as spreading tissue inside a tropical vine. A parasite, it steals food from the vine because it cannot photosynthesize for itself.

Hotbed of nutrition

A rainforest floor in Borneo is a busy community of living things, all striving for nourishment. While green-leaved plants make the food upon which, ultimately, everything else depends, a multitude of predators, parasites, and decomposers are fed by living prey and an abundance of dead matter.

Mountain tree shrews nourish the pitchers with their droppings—and are rewarded with a lick of sweet nectar.

Insectivorous plants

Where the soil is low in certain minerals, some plants seek other sources of food. The leaves of pitcher plants develop into vessels that contain pools of fluid for digesting drowning insects and even the droppings of occasional mammals.

Saprophytes

Toadstools and other fungi are saprophytes—meaning that they absorb the liquified remains of dead matter. They are made up of microscopic filaments, called hyphae, that penetrate the soil and cling to dead matter, simultaneously releasing digestive juices and soaking up the digested products.

Soil contains dead matter, which releases minerals into the ground as it decomposes.

Bacteria

Most kinds of bacteria digest dead matter, driving the process of decomposition. Others process the chemical energy in minerals to make their own food and, in doing so, release nitrates—an important source of nitrogen sucked up by plant roots.

Detritivores

A forest floor is littered with organic detritus (waste), such as dead leaves. This provides abundant food for detritivores, such as giant blue earthworms, that have the digestive systems to cope with this tough material.

Many detritus-eating animals burrow in soil, where they are surrounded by their food.

